Complex Multiplication and Cancellation in Elliptic Trace Heights

10 November 2026

Abstract

For the curve $E_D : y^2 = x^3 - Dx$ over a number field K not containing i, adjoining i doubles the Mordell–Weil rank. If $P \in E_D(K)$ is non-torsion, the point $C = (1+i)P$ generates an infinite cyclic subgroup whose nonidentity abscissas all have relative trace zero, while their heights grow quadratically. We also give four pairwise distinct doubled points, with distinct abscissas, whose abscissa sum is zero and whose heights are unbounded. The curve with $D = 11 - 6\sqrt{2}$ provides an explicit example over $\mathbb{Q}(\sqrt{2}, i)$; a two-prime reduction certificate proves that the displayed point has infinite order. These results obstruct an unrestricted transfer of the particular elliptic rank and height inputs to a complex field. They do not give a counterexample to Diophantine definability of a ring of integers in its fraction field.

1 The curve and its endomorphisms

Let K be a number field with $i \notin K$, let $D \in K^*$, and put

$$E_D : y^2 = x^3 - Dx, \qquad L = K(i).$$

In particular, K may be any number field with a real embedding. The curve is nonsingular, with discriminant $64D^3$. Let σ be the nonidentity element of $\mathrm{Gal}(L/K)$. Write O for the identity of E_D. The automorphism

$$\iota(O) = O, \qquad \iota(x, y) = (-x, iy) \tag{1}$$

is defined over L. It preserves the curve equation and the chord-and-tangent law, and satisfies

$$\iota^2 = [-1], \qquad \sigma(\iota Q) = -\iota(\sigma Q) \quad (Q \in E_D(L)). \tag{2}$$

The notation $(1+\iota)P$ means $P + \iota P$, and similarly for $1 - \iota$. Thus

$$(1-\iota)(1+\iota) = [2]. \tag{3}$$

These identities are identities of elliptic endomorphisms, not statements about a chosen generator of a Mordell–Weil group.

Lemma 1.1. *If $P = (x, y)$ and $x \neq 0$, then*

$$\begin{aligned} x((1+\iota)P) &= -\frac{i(x^2 - D)}{2x}, \\ y((1+\iota)P) &= -\frac{(1+i)y(x^2 + D)}{4x^2}. \end{aligned} \tag{4}$$

The endomorphism $1 + \iota$ is a degree-two isogeny with kernel $\{O, (0,0)\}$.

Proof. The line through (x, y) and $(-x, iy)$ has slope $(1 - i)y/(2x)$. Its third-intersection formula gives

$$x((1 + \iota)P) = \left(\frac{(1 - i)y}{2x}\right)^2 = -\frac{i(x^2 - D)}{2x}.$$

Substituting into the ordinate formula gives the second identity. If $(1+\iota)Q = O$, then $\iota Q = -Q$. For an affine point this implies $-x(Q) = x(Q)$, hence $x(Q) = 0$ and $Q = (0, 0)$. Conversely $(0, 0)$ lies in the kernel. In characteristic zero a nonzero elliptic endomorphism is separable, so its degree is the number of geometric kernel points, namely two [3, Chapter III, Sections 4 and 6]. □

2 Exact rank growth

Theorem 2.1. *For every K, D, L as above,*

$$\operatorname{rank} E_D(L) = 2 \operatorname{rank} E_D(K).$$

More precisely,

$$2E_D(L) \subseteq E_D(K) + \iota E_D(K), \tag{5}$$

and $E_D(K) \cap \iota E_D(K)$ consists of points killed by two.

Proof. For $Q \in E_D(L)$, put

$$A = Q + \sigma Q, \qquad B = -\iota(Q - \sigma Q).$$

Both points are fixed by σ: the assertion for B uses (2). Thus $A, B \in E_D(K)$, and

$$2Q = A + \iota B.$$

This proves (5). If $T \in E_D(K) \cap \iota E_D(K)$, write $T = \iota U$ with $U \in E_D(K)$. Then $T = \sigma T = -\iota U = -T$, so $2T = O$. The sum $E_D(K) + \iota E_D(K)$ therefore has rank $2 \operatorname{rank} E_D(K)$. It has finite index in $E_D(L)$, since it contains $2E_D(L)$ and the Mordell–Weil group is finitely generated. The ranks are equal, proving the formula. □

In particular, if $E_D(K)$ has positive rank, the same curve cannot retain that rank after adjoining i. If its rank over K is one, its rank over $K(i)$ is two. Selecting a different elliptic curve is a separate possibility: Zywina's theorem provides curves of rank one stable in every prescribed quadratic extension [4, Theorem 1.2]. The obstruction here concerns the specified family with $j = 1728$.

3 An infinite cyclic subgroup with zero trace

Let h denote the absolute logarithmic Weil height, with $h(0) = 0$. Normalize the canonical height on E_D by

$$\widehat{h}(Q) = \frac{1}{2} \lim_{m \to \infty} 4^{-m} h(x([2^m]Q)).$$

We use its quadraticity, Galois invariance, parallelogram law, and positivity away from torsion, together with

$$h(x(Q)) = 2\widehat{h}(Q) + O_{E_D}(1). \tag{6}$$

These are the standard canonical-height properties [3, Chapter VIII, Section 9].

Theorem 3.1. *Let $P \in E_D(K)$ have infinite order and put $C = (1+\iota)P$. Then C has infinite order. For every nonzero integer n,*

$$\mathrm{Tr}_{L/K}(x([n]C)) = 0, \qquad \mathrm{Tr}_{L/\mathbb{Q}}(x([n]C)) = 0, \tag{7}$$

whereas

$$h(x([n]C)) = 4n^2\widehat{h}(P) + O_{E_D}(1). \tag{8}$$

In particular, even on the cyclic subgroup $\mathbb{Z}C$, the height of an abscissa cannot be bounded by any function of the height of its relative or absolute trace that is finite at zero. The same conclusion holds on $2\mathbb{Z}C \subseteq 2E_D(L)$.

Proof. Equation (3) sends C to $[2]P$, so C is non-torsion. Since $\sigma P = P$,

$$\sigma C = (1-\iota)P = -\iota(1+\iota)P = -\iota C.$$

For $Q = [n]C$ this gives

$$\sigma(x(Q)) = x(\sigma Q) = x(-\iota Q) = -x(Q).$$

The points are non-torsion, so their abscissas are finite. This proves the relative trace identity; transitivity of trace proves the absolute one.

The equality $x(\iota Q) = -x(Q)$ and the height definition give $\widehat{h}(\iota Q) = \widehat{h}(Q)$. Galois invariance gives $\widehat{h}(P+\iota P) = \widehat{h}(P-\iota P)$. The parallelogram law therefore yields

$$2\widehat{h}(C) = \widehat{h}(P+\iota P) + \widehat{h}(P-\iota P) = 2\widehat{h}(P) + 2\widehat{h}(\iota P) = 4\widehat{h}(P).$$

Quadraticity and (6) prove (8). Its leading coefficient is positive, while both trace heights are identically zero. Taking only even n proves the last assertion. □

4 Four distinct doubled points with cancelling abscissas

Theorem 4.1. *For $n \geq 1$, set*

$$Q_{1,n} = [2n]P, \quad Q_{2,n} = \iota[2n]P, \quad Q_{3,n} = [4n]P, \quad Q_{4,n} = \iota[4n]P.$$

These are four pairwise distinct non-torsion points in $2E_D(L)$. Their abscissas are pairwise distinct and satisfy

$$\sum_{j=1}^{4} x(Q_{j,n}) = 0. \tag{9}$$

Nevertheless

$$\min_{1\leq j\leq 4} h(x(Q_{j,n})) \longrightarrow \infty \quad \text{as } n \longrightarrow \infty.$$

Consequently there are no constants A, B for which

$$\max_j h(x(Q_j)) \leq A\, h\left(\sum_j x(Q_j)\right) + B$$

holds for all four-tuples of distinct non-torsion doubled points of $E_D(L)$, even with pairwise distinct abscissas.

Proof. All four points are doubles and non-torsion. Their abscissas cancel in pairs by (1). If $aP = \iota bP$ for nonzero integers a, b, applying $[a] + [b]\iota$ gives $[a^2+b^2]P = O$, a contradiction. The same argument permits either sign on the right. Equal abscissas on an elliptic curve mean equal or opposite points; thus no abscissa from one of the two ι-directions can equal one from the other. Within either direction, equality would imply $[2n]P = \pm[4n]P$, again impossible. This proves both distinctness claims. Finally (6) gives

$$h(x([2n]P)) = 8n^2\widehat{h}(P) + O_{E_D}(1), \qquad h(x([4n]P)) = 32n^2\widehat{h}(P) + O_{E_D}(1).$$

The ι-images have the same heights, proving the assertions. □

5 An explicit instance

Take

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = 11 - 6\sqrt{2},$$
$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right) = [2]R_0.$$

Both curve identities and the doubling identity are exact. The following short verification of non-torsion is independent of any rank or Diophantine-definability input.

Lemma 5.1. *The displayed point P has infinite order.*

Proof. Use the degree-one primes of K specified by $\sqrt{2} \equiv 6 \pmod{17}$ and $\sqrt{2} \equiv 5 \pmod{23}$. Both are good-reduction primes. Direct finite-field calculation gives

p	$D \bmod p$	$P \bmod p$	$\#E_D(\mathbb{F}_p)$	$\operatorname{ord}(P \bmod p)$
17	9	$(1, 14)$	20	5
23	4	$(4, 5)$	24	3.

If P were torsion, its reduction of order five at the first prime would force its order to be divisible by five. A point of order five would then belong to $E_D(K)$. Reduction at the second prime is injective on five-primary torsion [3, Chapter VII, Section 3], but its group has order 24. This is impossible. □

Theorems 2.1, 3.1 and 4.1 now apply over $\mathbb{Q}(\sqrt{2}, i)$ without additional arithmetic hypotheses. The point $C = (1 + \iota)P$ is itself a double because $P = [2]R_0$. Thus all the trace-cancelling multiples in this explicit instance already lie in $2E_D(\mathbb{Q}(\sqrt{2}, i))$.

6 Scope for definitions over number fields

A ring of integers $\mathcal{O}_L$ is Diophantine in L if it is the projection of a polynomial zero set with all witnesses ranging over L. None of the preceding theorems disproves the existence of such a definition. They rule out transferring the particular rank-one and elliptic trace-height assertions to unrestricted points of the larger field.

The cancellation in Theorem 3.1 is an explicit Galois relation among the points. It violates the independence hypothesis required by the aggregate trace-height theorem [1]. The cancellation in Theorem 4.1 shows why a sum-height argument relying on positive abscissas at real embeddings requires a different justification over a field containing i. The rank formula shows that even the condition of being a double does not retain the original cyclic group.

The isogeny-multiplicity argument for modular contact heights [2] is separate. Its height averaging and dimension-two isogeny estimates are formulated over arbitrary number fields; they do not require real embeddings. The obstruction identified here is in the elliptic group and trace or sum encoding. A contact-height theorem for algebraic parameters would still require the appropriate modular realization and its stated uniform estimates, rather than an unrestricted change of its rational parameter field.

7 Exact verification

The standard-library program `computation/number_fields/check_trace_cancellation.py` checks the coordinate identities in the universal ring

$$\mathbb{Q}[D, x, x^{-1}, y, i]/(y^2 - x^3 + Dx,\ i^2 + 1).$$

It verifies the automorphism, chord formulas, both coordinates of $(1-\iota)(1+\iota)P = [2]P$ after clearing denominators, conjugation, and the group decomposition used for rank growth. It also verifies the displayed point and its doubling, and counts the two finite reduction groups by two independent elementary methods. Their point orders give the non-torsion certificate above.

Several exact multiples are included as supplementary checks of the trace and four-point identities. The general identities and height-growth conclusions follow from the proofs, not from those finite samples. The output is `computation/number_fields/trace-cancellation-certificate.json`.

References

[1] *Aggregate trace heights of elliptic multiples*, companion manuscript, 8 November 2026. `papers/trace_heights/aggregate_trace_height_20261108_v9.tex`.

[2] *Multiplicity of modular factors and quadratic contact-height bounds*, companion manuscript, 10 November 2026. `papers/contact_heights/multiplicity_and_contact_heights_20261110_v19.tex`.

[3] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd edition, Graduate Texts in Mathematics 106, Springer, 2009; Chapters III, VII and VIII. https://doi.org/10.1007/978-0-387-09494-6.

[4] D. Zywina, *Rank one elliptic curves and rank stability*, arXiv:2505.16960, Theorems 1.1 and 1.2. https://davidzywina.github.io/papers/Rank1WithStability.pdf.